

MATH 250: TOPOLOGY I BONUS PROBLEMS

FALL 2026

Cite all sources, including people, books, papers, websites, and AI tools. Write up your answers in your own words.

Unless otherwise specified, $\mathbf{R}^n$ has the analytic topology for all n , any subset of $\mathbf{R}^n$ has the subspace topology, and any product of topological spaces has the product topology. We write S^{n-1} for the Euclidean unit $(n-1)$ -sphere in $\mathbf{R}^n$. For convenience, we write p_θ to denote the point $(\cos 2\pi\theta, \sin 2\pi\theta) \in S^1$.

1. DUE WEDNESDAY, SEPTEMBER 23

Problem 1. Recall that an *embedding* is an injective continuous map. Give an explicit embedding of the infinite cylinder $S^1 \times \mathbf{R}$ into $\mathbf{R}^2$.

Problem 2. Fix integers a, b . Let $K_{a,b}$ be the set of pairs of complex numbers $(z, w) \in \mathbf{C}^2$ such that $|z| = |w| = 1$ and

$$w^b = z^a.$$

Explain how $K_{a,b}$ can be used to construct an explicit embedding of S^1 into $S^1 \times S^1$ when a and b are coprime. What happens when a and b are not coprime?

Problem 3. Let G be the set of 2×2 complex matrices g such that $\bar{g}^t g = I$, where I is the identity matrix and $\bar{g}^t$ is the complex conjugate of the transpose of g .

- (1) Show that G forms a group.
- (2) Identifying $\mathbf{C}$ with $\mathbf{R}^2$, show that G is homeomorphic to S^3 .
- (3) What is the usual name for G in the literature?

Problem 4. An *inverse system* consists of:

- A poset I (say, with partial order $\preceq$).
- A collection of sets $\{X_i\}_{i \in I}$.
- A collection of maps $\{\phi_{i,j} : X_j \rightarrow X_i\}_{i \preceq j}$, such that for all $i, j, k \in I$ with $i \preceq j \preceq k$, we have $\phi_{i,k} = \phi_{i,j} \circ \phi_{j,k}$.

Show that the following data form an inverse system:

- The poset is the collection I of bounded sets in $\mathbf{R}$ under $\subseteq$.
- The sets are $X_A = \{\text{functions from } A \text{ to } \mathbf{R}\}$ for $A \in I$.
- The map $\phi_{A,B} : X_B \rightarrow X_A$ is given by $\phi_{A,B}(f) = f|_A$, where $|_A$ means restriction of domain, for all $A, B \in \mathcal{B}$ with $A \subseteq B$.

Problem 5. Given an inverse system with sets $\{X_i\}_{i \in I}$ and maps $\{\phi_{i,j} : X_j \rightarrow X_i\}_{i \preceq j}$, as above, we define the *inverse limit* $\varprojlim_i X_i$ to be the set

$$\varprojlim_i X_i = \left\{ (x_i)_i \in \prod_{i \in I} X_i \mid \phi_{i,j}(x_j) = x_i \text{ for all } i, j \in I \text{ with } i \preceq j \right\}.$$

Let $X_{\mathbf{R}} = \{\text{functions from } \mathbf{R} \text{ to } \mathbf{R}\}$.

- (1) Show that for the inverse system in Problem 4, the map $X_{\mathbf{R}} \rightarrow \varprojlim_A X_A$ defined by $f \mapsto \{f|_A\}_{A \in I}$ is a bijection.
- (2) If we replace “function” with “bounded function” in the definitions of X_A and $X_{\mathbf{R}}$, does (1) still hold?

Problem 6. Fix a commutative ring R and an ideal $N \subseteq R$.

- (1) Show that the following data form an inverse system:
 - The poset is $\mathbf{Z}_{>0}$ under $\leq$.
 - The sets are R/N^i for $i \in \mathbf{Z}_{>0}$.
 - The map $\phi_{i,j} : R/N^j \rightarrow R/N^i$ is the quotient modulo N^i for all $i, j \in \mathbf{Z}_{>0}$ with $i \leq j$.
- (2) The inverse limit $\varprojlim_i R/N^i$ is called the *completion* of R with respect to N . Show that it forms a subring of $\prod_i R/N^i$. What is its additive identity 0? What is its multiplicative identity 1?
- (3) Show that the diagonal map $R \rightarrow \prod_i R/N^i$ factors through a map

$$R \rightarrow \varprojlim_i R/N^i.$$

When is the latter injective?

Problem 7. Take $R = \mathbf{Z}$ and $N = p\mathbf{Z}$ for a positive integer p in Problem 6. Here, the inverse limit is called the *ring of p -adic integers* and denoted $\mathbf{Z}_p$.

For any nonzero $\alpha = (\alpha_i)_i \in \mathbf{Z}_p$, let the largest index $i \geq 0$ such that $\alpha_j = 0$ for all $j < i$ be denoted $v(\alpha)$. Let $|0| = 0$, and for nonzero α , let $|\alpha| = p^{-v(\alpha)}$.

- (1) Show that $d : \mathbf{Z}_p \times \mathbf{Z}_p \rightarrow \mathbf{R}$ defined by

$$d(\alpha, \beta) = |\beta - \alpha|$$

is a metric. It is called the *p -adic metric*.

- (2) Show that here, the map in Problem 6(3) is injective, so $\mathbf{Z}$ forms a subring of $\mathbf{Z}_p$. But in the p -adic metric, $\mathbf{Z}_p$ is bounded, so the Archimedean property fails.
- (3) In the p -adic metric, is $\mathbf{Z}$ dense in $\mathbf{Z}_p$?